

Pokhara University Faculty of Science and Technology	
Course Code: CMP 335(3 Credit)	Full Marks: 100
Course Title: Data Communication (3-0-2)	Pass Mark: 45
Nature of the Course: Theory and Practice	Total Lectures: 45 hours
Level: Bachelor/ Year: III/ Semester: III	Program: Bachelor in Computer Engineering

4. Course Contents
Unit 1: Introduction (2 hrs)
• Evolution of Data Communication systems • Analog and Digital Data Transmission, Data Communication Terminology • Standards Organizations, Applications
Unit 2: Data Transmission (3 hrs)
• Data Transmission Techniques - Parallel Transmission, Serial Transmission (Synchronous, Asynchronous and Isochronous Communication), Modes of Data Transmission • Line Configuration, Bit Rate/ Baud rate, Transmission Channel, Data Rate Limits - Shannon Capacity Theorem and Nyquist Bit Rate, RS-232C (DTE-DCE, DTE-DTE)
Unit 3: Signals and Systems (6 hrs)
• Signals and their classification: Periodic and non-periodic signals; Deterministic and Random signals; Energy and Power signals; Continuous and Discrete time signals • Basic Elementary Signals - Unit Step Signal, Ramp Signal, Impulse Signal, Sinusoidal Signal, Signum Signal • System - Continuous and Discrete time system • Basic system properties: Linearity, Causality, Stability, Static & Dynamic, and Time Invariance, Introduction to LTI System
Unit 4: Overview of Data Communication Networking and Protocols (4 hrs)
• Network Types, Topology • OSI layers and Functions, TCP/IP layer, Local Area Networks (LAN) Architecture, LLC/MAC & Routing • IEEE Standards, Ethernet (Aloha, CSMA), Wide Area Networks (WAN): X.25, Frame Relay, ATM
Unit 5: Transmission Media (5 hrs)

• Electromagnetic Spectrum for Telecommunication • Type of Propagation • Guided Transmission Media: Twisted Pair Cable, Coaxial Cable, Optical Fiber, Characteristics of Unguided Communication Bands, Antennas • Unguided Transmission Media: Terrestrial Microwave, Satellite Communication, VSAT, and Cellular Telephony
Unit 6: Impairments, Error handling and Compression Techniques (6 hrs)
• Attenuation & Distortion, Delay Distortion, Noise & Types, interference, crosstalk • Types of error & its Detection and Correction Methods • Data Compression, Lossless Compression - Run Length Coding, Dictionary Coding and Huffman Coding • Lossy Compression - Predictive Coding and Transform Coding
Unit 7: Data Link Control and Protocol (5 hrs)
• Framing • Flow Control: Stop - & - Wait, Sliding Window, Error Control: Automatic Repeat Request (ARQ), Stop-and Wait ARQ, Sliding Window (ARQ) • HDLC protocol • Point-to-Point protocol
Unit 8: Multiplexing & Switching (5 hrs)
• Multiplexing types and Application • Multiplexing Vs Non-Multiplexing • The Telephone System: Analog services and its Hierarchy • Digital services and Hierarchy Circuit Switching, Packet Switching, Message Switching, and Private Branch Exchange
Unit 9: Data Encoding & Modulation (9 hrs)
• Line coding – Unipolar, Polar and Bipolar signaling, • Digital Modulation Techniques - Amplitude, Frequency, and Phase Shift Keying. • Analog to Digital Conversion - Pulse Code and Delta Modulation. • Analog Modulation Techniques -Amplitude, Frequency, and Phase Modulation • Multilevel Modulation-QPSK, QAM • Introduction to Modem

Course Code: MTH 210
 Course title: Calculus II (3-2-0)
 Nature of the Course: Theory
 Level: Bachelor

Full Marks: 100
 Pass Marks: 45
 Total Lectures: 45 hours
 Program: BE

Specific objectives	Contents
Evaluate multiple integrals	Unit I: Multiple Integrals (6 Hours) 1.1 Introduction 1.2 Double integrals in Cartesian and polar form, Fubini's theorem (statement only), change of order of integration, change of variable from in double integral Jacobian matrix and reduction into Polar. 1.3 Triple integrals in Cartesian form and Dirichlet's Integral, use of cylindrical and spherical coordinates to evaluate triple integral. 1.4 Application of double and triple integrals to find Area and volume.
Analyze the concept of solution of differential equations in terms of infinite series as power series.	Unit II: Series Solution of Differential Equations and Special Functions (6 Hours) 2.1 Power series method of solution of differential equations. 2.2 Legendre's Equation, Legendre's polynomials $P_n(x)$ of . Graph of $P_1(x), P_2(x), P_3(x)$. 2.3 Frobenius method. Bessel's equation, Bessel's function $J_v(x)$ and its properties. Graph of $J_v(x)$ for $v=1 \square 2$
Apply the integral transform in solving practical problems	Unit III: Laplace Transform and Its Application (8 Hours) 3.1 Laplace Transform (LT), Inverse LT, Linearity of LT, LT of elementary functions, inverses and first shifting (s-shifting) theorem. Existence theorem of Laplace transform (without proof) and uniqueness. 3.2 Transform of Derivative and Integrals of a function. 3.3 Differentiation and Integration of Laplace transform. 3.4 Unit step function, periodic function and LT, second shifting (t-shifting) theorem. 3.5 Convolution theorem and its application to find inverse. 3.6 Application of Laplace transform to find the solutions of ordinary differential equations (IVP).
Solve higher dimensional (multivariable) calculus problems	Unit IV: Advanced Vector Calculus (15 Hours) 4.1 Differentiation of vector function of scalar variable. 4.2 Point functions, Gradient, directional derivative, divergence and curl with properties (without proof) 4.3 Line integral with physical interpretation and evaluation of line integrals on various path 4.4 Line integral, potential function and independence of path 4.5 Green's theorem in plane (without proof) and its various

	applications 4.6 Surface integral and evaluation of surface integrals 4.7 Stoke's theorem (without proof) and its applications 4.8 Gauss Divergence theorem (without proof) and its applications.
Illustrate periodic functions of practical importance by infinite trigonometric series	Unit V: Fourier Series (5 Hours) 5.1 Periodic Functions, odd and even functions 5.2 Fourier series of 2π periodic functions in the interval $(\alpha, \alpha + 2\pi)$. 5.3 Fourier series of $2l$ periodic functions. 5.4 Fourier series of odd and even functions, sine and cosine series
Interpret physical phenomenon by partial differential equations	Unit VII: Partial Differential Equations (5 Hours) 7.1 Introduction 7.2 Linear constant coefficient equation 7.3 Applications in conservation laws, the breaking time, shock waves, nonlinear advection equations, and traffic flow.

Course Code: CMP 234
 Course title: Computer Graphics (3-1-2)
 Nature of the Course: Theory and Practical

Full Marks: 100
 Pass Marks: 45
 Total Lectures: 45 hours

Specific Objectives	Contents
Provides a brief introduction of the field of Computer Graphics and familiarize students with basic graphics hardware and software. It intends to enhance the understanding of the technology used for producing graphical output on display devices.	Unit 1: Overview of Computer Graphics and Graphics Systems (7 hrs) 1.1 Introduction, Applications and Recent trends of Computer Graphics 1.2 Interactive Input Devices 1.3 Display Devices, Color Monitors, Hard Copy Devices 1.4 Raster and Random Scan Systems and Architectures 1.5 Video Controller 1.6 Use of Digital to Analog Converter and Frame Buffer Organization 1.7 Graphics Software, Modern Graphics Hardware (GPU)
Learn about various scan conversion and area filling algorithms. Students will be able to implement these scan conversion algorithms using a programming language. This chapter helps students learn how algorithms are used to generate output primitives and fill them with specified intensities.	Unit 2: Scan Conversion Algorithms (7 hrs) 2.1 Scan Conversion 2.2 Line Drawing Algorithms 2.2.1 Digital Differential Analyzer 2.3.2 Bresenham's Algorithm 2.3 Circle Generation Algorithm 2.4 Ellipse Generation Algorithm 2.5 Filled Area Primitives 2.5.1 Scan Line Polygon Fill Algorithm 2.5.2 Boundary Fill Algorithm, Flood Fill Algorithm
Get students well acquainted with the two-dimensional geometric transformations and viewing Able to make use of various transformation operations and viewing operations to reposition and resize objects in two dimensional scene Learn about the operations used in viewing routines that convert a world coordinate scene description to a display for an output device.	Unit 3: Two Dimensional Geometric Transformations and Viewing (8 hrs) 3.1 Two Dimensional Transformations 3.1.1 Translation, Rotation, Scaling, Shearing, Reflection 3.2 Matrix Representations of Transformations 3.3 Homogeneous Coordinate System 3.4 Composite Transformations 3.5 Windowing Concepts, Two Dimensional Viewing Pipeline 3.6 Window to Viewport Transformation 3.7 Line Clipping Algorithm: Cohen-Sutherland 3.8 Polygon Clipping: Sutherland-Hodgeman

Learn about the extensions of two-dimensional methods for geometric transformation in three dimensions. Helps them apply the knowledge of three dimensional rendering and transformations for representing objects more accurately	Unit 4: Graphics in Three Dimensions (8 hrs) 4.1 Three Dimensional Coordinate Systems 4.2 Three Dimensional Transformations 4.2.1 Translation, Rotation, Scaling, Reflection 4.3 Three Dimensional Representations 4.3.1 Polygon Surfaces 4.3.2 Cubic Spline and Beizer Curve 4.3.3 Non-Planer Surface: Bezier Surface 4.3.4 Fractal Geometry Method 4.4 Three Dimensional Viewing Transformation and Pipeline 4.5 Projections 4.5.1 Parallel Projection 4.5.2 Perspective Projection 4.6 Clipping in Three Dimensions
Get acquainted with the techniques used for bringing visual realism in computer generated graphical output primitives. Familiarize with visible surface detection techniques for creating realistic displays and in computing	Unit 5: Visual Realism (7 hrs) 5.1 Hidden Surfaces and Hidden Surface Removal Approaches 5.1.1 Back-Face Detection 5.1.2 Depth Buffer Method 5.1.3 A- buffer method 5.1.4 Scan Line Method 5.1.5 Depth Sorting Method
light intensities of surfaces using lighting or shading models for generating realistic views	5.2 Illumination and Shading Methods 5.2.1 Illumination Theory and Models 5.2.1.1 Ambient Light 5.2.1.2 Diffuse Reflection 5.2.1.3 Specular Reflection (Phong Model) 5.2.2 Polygon Surface Shading Methods 5.2.2.1 Constant Shading (Flat Shading) 5.2.2.2 Gouraud Shading 5.2.2.3 Phong Shading 5.2.2.4 Fast Phong Shading 5.3 Color Models 5.3.1 RGB 5.3.2 CMYK
Gain a good understanding of the use of graphical standards necessary for achieving software portability. Become familiar with the basic structure or format of a graphical file and know the ways of visually representing data sets. Learn about writing graphical programs that can be ported to multiple hardware platforms without extensive rewriting of the programs through the use of graphical standards.	Unit 6: Graphical Standards (4 Hrs) 6.1 Need for Machine Independent Graphical Languages 6.2 Graphical Standards: PHIGS, GKS 6.3 Graphics Software Standards and Language Binding 6.4 Overview of Graphics File Formats 6.5 Visualization of Data Sets

Learn about the library functions provided in OpenGL for graphical rendering and performing geometric transformations. Enable students to write programs using functions in OpenGL library for drawing graphical output primitives and performing basic transformations	Unit 7: Introduction to OpenGL (4 Hrs) 7.1 Overview and Basic Architecture of OpenGL 7.2 GL Related Libraries 7.2.1 Drawing Basic Output Primitives and Polygons 7.2.2 Call Back Functions and Input Handling 7.2.3 Basic Transformations 7.2.4 Projections and Lighting
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Course Code: CMP 222
 Course Title: Database Management System (3-1-3)
 Nature of the Course: Theory and Practical
 Level: Bachelor

Full Marks: 100
 Pass Marks: 45
 Total Lectures: 45hours
 Program: BE

1. Contents in Detail

Specific Objectives	Contents
Familiarize with basic concepts of database systems, and its architecture.	Unit 1: Introduction [4 hrs] 1. Database Management System purpose and applications, 2. Database Systems vs File Systems 3. View of Data- Data Abstraction (Physical, logical and view level, Data Independence) 4. Instances and Schemas, 5. Database Languages (DDL, DML and DCL) 6. Database and Application Architecture- Database System Architecture and Database Application Architecture (two-tier and three-tier)
- Familiarize with ER Model and Relational Model and relational algebra. - Design the relational database using ER model.	Unit 2: ER and Relational Model [8 hours] 1. Introduction to ER Model: 2. Entity sets, attributes and values, Relationship sets- participation, entity's role, descriptive attributes, degree of relationship set, Mapping Cardinalities, Attributes- simple, composite, single-valued, multi- valued, derived, Entity-Relationship (ER) Diagram, Specialization, Generalization, and Aggregation 3. Key- Super key, Candidate key and Primary key, Strong and Weak Entity Sets 4. Introduction to Relational Model 5. Reducing ER diagrams to Relational Schema 6. Structure of Relational Databases, Database Schema, Keys, Schema Diagrams 7. Relational Algebra
Implement and write DDL and DML queries in the SQL.	Unit 3: Structured Query Language [5 hours] 1. Structured Query Language (SQL)- SQL DDL and DML, 2. Basic Structure of SQL Queries, DDL queries, Basic Operations (Rename, String, Attribute Specification in the select clause, order by, where-clause), Set Operations, Null values, Aggregate Functions, 3. Nested Queries, 4. Join Expressions (Natural Join, Join Conditions, Outer Joins), 5. Modification of Database (delete, insert into, update), 6. Views, Stored Procedures

• Apply the integrity constraints to implement database securities. • Normalize the database to a defined normal form.	Unit 4: Relational Database Design [6 hrs] 1. Integrity constraints- Domain Constraints, Entity Integrity Constrains, Referential Integrity Constraints, Assertions and Triggers 2. Features of Good Relational Designs 3. Functional dependencies and Armstrong's Axioms 4. Closure of a Set of Functional Dependencies and Closure of Attribute Sets 5. Database Normalization and Normal Forms- 1NF, 2NF, 3NF and BCNF 6. Denormalization for Performance
• Understand the need of database security. • Understand the role of access control, authorization, views and encryption mechanism to provide database security.	Unit 5: Security [3 hrs] 1. Security and integrity violations 2. Access control 3. Authorization 4. Security and Views
underderstand the mechanism of query accessing and need of query optimization.	Unit 6 : Query Processing and Optimization [5 hrs] 1. Introduction to Query Processing 2. Equivalence of Expressions 3. Query Cost Estimation 4. Query Optimization 5. Query evaluation and execution plan
• Familiarize with different file organization methods.	Unit 7 : Storage Management and Indexing [3 hrs] 1. File Organization- fixed length records, variable length records 2. Organization of Records in File- Heap, Sequential and Indexed sequential file organizations. 3. B+ Tree Index Files 4. Hash Indices
• Understand the concepts of transactions, need of atomicity, durability and isolation, serial schedule. • Familiarize with the need of Concurrency control.	Unit 8 : Transactions and Concurrency Control [4 hrs] 1. Transaction Concepts 2. Transaction Model and State Diagram 3. ACID properties of transaction 4. Serializability- conflict and view serializability 5. SQL Standard Isolation Levels 6. Concurrency Control- Lock-Based Protocols and Graph-Based Protocols

Understand the recovery algorithms and techniques to protect and recover the data from various failures.	Unit 9 : Crash Recovery [3 hrs] 1. Failure classification 2. Recovery and Atomicity- log records, database modification, concurrency control and recovery, transaction commit, Redo and Undo Transactions using Log, Check Points, 3. Recovery Algorithm Using Log Records- Transaction Rollback, Recovery After a System Crash, Optimizing Commit Processing 4. High Availability Using Remote Backup System
Understand the concepts of NoSQL, ODBMS, Distributed Databases	Unit 10 : Emerging Trend in Databases [4 hrs] 1. NoSQL Databases- Characteristics, Categories, Advantages 2. Object Oriented Database and ORM 3. Distributed Databases- 4. Distributed Ledged Technology: Blockchain, Cryptocurrency, Blockchain Properties

Course Code: CMP 232
 Course title: Operating Systems (3-1-2)
 Nature of the Course: Theory and Practical
 Level: Bachelor

Full Marks: 100
 Pass Marks: 45
 Total Lectures: 45hr
 Program: BE

Specific Objectives	Contents
- Familiarize with basic concepts of Operating systems, and its architecture. - Understanding the successive evolution of operating system	Unit 1: Introduction [4 hrs.] 1.1 Concept and function of operating systems 1.2 Operating systems concept and functionality - Processes, Files, System Calls, The Shell 1.3 Operating System Structure - Monolithic System, Layered, Virtual Machines, Client-server. Microkernel and exokernel 1.4 Types and Evolution of Operating Systems
- Familiarize with Task, Process and threads - Implement of resource allocation techniques - Understanding the mutual exclusion for resource utilization	Unit 2: Processes and Threads [10 hours] 2.1 Process - Definition, states and transition diagram - Process control block (PCB) - Concurrent and Parallel processes 2.2 Interprocess Communication and Synchronization - Introduction, Race Condition, Critical Regions and condition, Avoiding critical region: Mutual Exclusion and Serializability, - Mutual exclusion conditions. mutual exclusion: disabling interrupts, lock variable, strict alteration (Dekker's algorithms, Peterson's algorithms), The TSL instruction, sleep and wakeup, producer and consumer problem - Types of mutual exclusion (Semaphore, Monitors, Bounded buffer, Message passing), - Classical IPC Problems (The Dining Philosophers problem, The Readers and Writers problem, The Sleeping Barber Problem) - Serializability: Locking Protocols and Time Stamp Protocols 2.3 Process Scheduling - Basic Concept, Type of scheduling (Preemptive scheduling, Non-preemptive scheduling, Batch, Interactive, Real time scheduling), - Scheduling criteria and performance analysis, scheduling algorithm with examples (First come first served, Shortest-job-first, Round-robin, Shortest process next, Shortest remaining time next, Real time, Priority fair share, guaranteed, Lottery scheduling) 2.3 Threads - Definitions of Threads, Types of thread

	process (Single and multithreaded process), Benefits of Multithreading - Threads Models (Many-to-one model, One-to-one Model, Many-to many model) - User Space and Kernel Space Threads, - Difference between Processes and Threads
• Conceptualize the role and working procedure of memory • Familiarizing with virtual memory management • Understanding the page replacement algorithms	Unit 3: Memory Management [10 hours] - Introduction: Storage organization, Memory hierarchy - Storage allocation, Contiguous versus non-contiguous store allocation, logical and physical memory - Fragmentation, fixed partition and variable partition for multiprogramming, Logical versus physical address space - Relocation and Protection - Memory management with swapping: Memory management with bitmaps and linked list, Memory management without swapping, - Contiguous-memory allocation: memory protection, memory allocation, Fragmentation (Inter fragmentation and external fragmentation) - Paging, Structure of page table: Hierarchical page table, Hashed page table, Inverted page table, Shared page table, - Virtual memory- Introduction, Paging, Page Table, Block mapping, Direct mapping, Translation Look Aside Buffers) Demand
	paging, Thrashing - Page replacement, Page replacement algorithms: First-in-first-out, Not recently used, Optimal page replacement, Second chance page replacement, Least Recently used, Clock page replacement, Working set page replacement, WS clock page replacement - Segmentation, Segmentation with paging - Coalescing and Compaction
• Conceptualization of Kernel and its role in system software	Unit 4: Kernel [2 hrs] 1. Introduction, Architecture of the Kernel, Types of Kernels 2. Context Switching (Kernel and User mode), Kernel implementation processes

• Understand the role of input/output devices • Understand the different approaches for optimal output	Unit 5: Input/Output Management [5 hrs] - Introduction, Interrupts Handlers - Principles of I/O Hardware (I/O Device, Device Controller, Memory Mapped I/O, Direct Memory Access) - Principles of I/O Software (Goals of I/O Software, Polled I/O verses Interrupt Driven I/O, Character User Interface and Graphical User Interface, Device Driver, Device Independent I/O Software, User -space I/O Software, - System Resources: Preemptable and Non-preemptable, Method of handling Deadlocks, Deadlock prevention, - Deadlock avoidance: Banker's Algorithm, Deadlock detection: Resource allocation graph, Recovery from Deadlock - Redundant Array of Inexpensive Disks - RAM Disks
• Understand the file and filing mechanism • Familiarize with directory and its management techniques	Unit 6: File Systems [3 hrs] - File and File Organization (Blocking and Buffering, File Descriptor, file Naming, File Structure, File Types, File Attributes, File Operations, File Access Methods) - Directories Management (Single-level directory systems, Hierarchical Directory systems, Path names, Directory operation) - Access Methods: Sequential, Direct, other access methods, Protection: Types of access, Access Control List, - File System Implementation: Contiguous allocation, Linked list allocation, linked list allocation using an Index nodes - Security and Multi-media files
• Familiarize with current trends of operating systems	Unit 7: New Trends in Operating System - Concept, character and role of - Real-time Operating System, - Distributed Operating System - Cloud Operating System - Mobile Operating System - Security issues and method of deployment - Memory wall and bottleneck for Operating system

Course Code: CMP 224
 Course title: Microprocessor and ALP (3-1-2)
 Nature of the Course: Theory & Practical
 Level: Bachelor

Full Marks: 100
 Pass Marks: 45
 Total Lectures: 45 hours
 Program: BE

Specific Objectives	Content
State and interpret block diagram of a Microprocessor based system and explain the function of each component, the line of communication Bus	Unit1: Introduction to Microprocessor (4 hrs.) 1.1 Microcomputer System 1.2 Microprocessor Operations 1.2.1 Microprocessor initiated operations 1.2.2 Internal operations 1.2.3 Peripheral initiated operations 1.3 Evolution of Microprocessor 1.4 Microcontroller and Microcontroller 1.5 Von Neumann and Harvard Architecture 1.6 Concept of Instruction Execution Cycle
- Describe the function of each block of the 8085 microprocessors - Explain the bus timings in fetching an instruction from memory - Explain the addressing Mode - Explain the function of data transfer instruction, Arithmetic instruction, logical and conditional and unconditional instruction. - Enable to write ALP. - Generation of control signal	Unit 2: 8085 Microprocessor Architecture(6) 2.1 Internal Architecture and features of 8085 Microprocessor 2.2 Addressing Modes of 8085 and its types 2.3 Instruction Cycle, Machine Cycle and T-State 2.4 Timing Diagram 2.4.1 Opcode fetch Cycle 2.4.2 Memory Read Cycle 2.4.3 Memory Write Cycle 2.4.4 I/O Read Cycle 2.4.5 I/O Write Cycle
- Describe the function of each block of the 8085 microprocessors - Explain the bus timings in fetching an instruction from memory - Explain the addressing Mode - Explain the function of data transfer instruction, Arithmetic instruction, logical and conditional and unconditional instruction. - Enable to write ALP. - Generation of control signal	Unit 3: Programming of 8085 microprocessor[5hrs] 3.1 Intel 8085 Instruction Set 3.2 Classification of Intel 8085 Instruction 3.3.1 Data Transfer Instruction, 3.3.2 Arithmetic Instruction 3.3.3 Logical Instruction, 3.3.4 Branching Instruction and 3.3.5 Stack, I/O and Machine control Instruction 3.4. Assembly Language Program 3.5.1 Format of an Assembly Language Instruction 3.6.2 Simple Programs with Arithmetic and Logical Operations, Conditions and Loops

• Interpret the block diagram of an 8086 Microprocessor and explain the function of each component. • Understand the concept of memory segmentation and any of physical addressing and effective addressing • Knowledge of real instructions and pseudo instructions • Enable to write advance programming in ALP • Analyze the concept of modular programming	Unit 4 8086 Microprocessor Architectural Details [10hrs] 4.1 Internal Architecture and features of 8086 Microprocessor 4.1.1 BIU and EU 4.1.2 Memory Segmentation 4.1.2.1 Segment and Offset Address 3.2 Intel 8086 Instruction Set 3.3 Classification of Intel 8085 Instruction 3.3.1 Data Transfer Instruction, 3.3.2 Arithmetic Instruction 3.3.3 Logical Instruction, 3.3.4 Branching Instruction 3.3.5 String manipulation Instruction 3.3.6 Machine control Instruction 3.5 Addressing Modes of 8086 and its types 3.5 Assembly Language program 3.5.1 Format of an assembly language Instruction 3.5.2 ALP Development Tools: Editor, Assembler, Linker, Debugger, Locator, Emulator 3.5.3 Assembler Directives 3.5.4 DOS Interrupt function(Keyboard and display services) 3.5.5 Simple Programs for Arithmetic, Logical, String Input/Output, Conditions and Loops 3.5.6 Stacks Procedures 3.5.7 Macros program design
• Interpret the block diagram of an 8086 Microprocessor and explain the function of each component. • Understand the concept of memory segmentation and any of physical addressing and effective addressing • Knowledge of real instructions and pseudo instructions • Enable to write advance programming in ALP • Analyze the concept of modular programming •	Unit 4 programming of 8086 microprocessor [10hrs] 3.1 Internal Architecture and features of 8085 Microprocessor 3.1.1 BIU and EU 3.1.2 Memory Segmentation 3.1.2.1 Segment and Offset Address 3.2 Intel 8086 Instruction Set 3.3 Classification of Intel 8085 Instruction 3.3.1 Data Transfer Instruction, 3.3.2 Arithmetic Instruction 3.3.3 Logical Instruction, 3.3.4 Branching Instruction 3.3.5 String manipulation Instruction 3.3.6 Machine control Instruction 3.6 Addressing Modes of 8086 and its types 3.5 Assembly Language program 3.5.1 Format of an assembly language Instruction 3.5.2 ALP Development Tools: Editor, Assembler, Linker, Debugger, Locator, Emulator 3.5.3 Assembler Directives 3.5.4 DOS Interrupt function(Keyboard and display services)

	3.5.5 Simple Programs for Arithmetic, Logical, String Input/Output, Conditions and Loops 3.5.6 Stacks Procedures 3.5.7 Macros program design
- Understand the concept on types of memory and its bus connection with microprocessor - Explain the difference between the peripheral mapped and memory-mapped technique - Interface and I/O device to a microcomputer by using logic gates and MSI Chip, such as decoders, latches and buffers. - Explain the Full Address decoding and partial decoding - List the element of the 8255 programmable peripheral interface (PPI) and explain its various operating modes and control word - Explain the process of the Direct Memory Access (DMA) and the function of various elements of the 8237.	Unit 4 Bus Structure, Memory, I/O and Interfacing [12 hrs.] 4.1 Bus Structure 4.1.1 Synchronous Bus 4.1.2 Asynchronous Bus 4.2 Memory Device Classification and Hierarchy 4.3 Interfacing I/O and Memory 4.3.1 Address Decoding 4.3.2 Unique and Non-Unique Address Decoding 4.3.3 I/O Mapped I/O and Memory mapped I/O 4.3.4 I/O Address Decoding (8085) 4.3.5 Memory Address Decoding (8085) 4.4 Parallel Interface 4.4.1 Modes: Simple I/O, Strobe I/O, Single Handshaking and Double Handshaking 4.4.2 Programmable Peripheral Interface (PPI) 4.4.2.1 Architecture of Intel (8255-PPI) 4.5 Serial Interface 4.5.1 Synchronous and Asynchronous Transmission 4.5.2 Programmable Communication Interface 4.5.2.1 USART (8251) 4.6 DMA and Intel 8237A DMA Controller
- Apply and analyze an Interrupt process and the difference between Maskable and Non Maskable Interrupt - Apply and analyze the instruction EI, DI, and RST and their function in the 8085-interrupt process. - Apply and analyze the concept of interrupt subroutine program - Apply and analyze the concept of interrupt processing - Apply and analyze the features of the PIC 8259	Unit 5 Interrupt Operation [8 Hrs.] 5.1 Interrupt, sources of interrupt and its Classification 5.2 Interrupt Processing Sequence in 8085/8086 5.3 Interrupt Instruction 5.3.1 Interrupts pins and priorities 5.3.2 Interrupt Service procedure 5.3.3 Interrupt Vector Table 5.3.4 Multiple Interrupt Handling 5.3.4.1 Polled and Vectored interrupt 5.3.5 Programmable Interrupt Controller (PIC) Intel (8259) and its Architecture